الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 9 · LESSONS 7-6 · 8-1 · 8-2 · 8-3

Weekly Quiz Preparation

Grade 11 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

Mr. Yousef Al Yousef · **Friday 30 October 2026**

1 Solving rational equations
حلّ المعادلات النسبية

Clear the fractions with the LCD, solve the resulting equation, and discard any solution that makes a denominator zero (an extraneous solution).

اضرب كل الحدود في المقسوم المشترك الأصغر، حلّ المعادلة، ثم استبعد أي حل يجعل المقام صفرًا (حل دخيل).

Steps الخطوات

1	**Factor the denominators and find the LCD**	حلّل المقامات واكتب الـ LCD
2	**Multiply every term by the LCD and cancel**	اضرب كل حد في الـ LCD واختزل
3	**Solve the equation that is left (often a quadratic)**	حلّ المعادلة الناتجة
4	**Check each answer: a zero denominator means extraneous**	تحقّق: المقام صفر يعني حلًّا دخيلًا

Worked example مثال محلول

Solve $\dfrac{2x}{x-3} - \dfrac{x^2+7x+6}{x^2-9} = \dfrac{6}{x+3}$.

$\text{LCD} = (x-3)(x+3)$	$x^2-9=(x-3)(x+3)$ حلّل المقام: $x^2-9=(x-3)(x+3)$
$2x(x+3) - (x^2+7x+6) = 6(x-3)$	multiply every term by the LCD اضرب كل حد في المضاعف المشترك الأصغر
$x^2-7x+12=0 \Rightarrow (x-3)(x-4)=0$	factor the quadratic حلّل المعادلة التربيعية
$x=3$ or $x=4$	both solve the new equation كلاهما يحقق المعادلة الجديدة
$x=3$ makes a denominator zero: extraneous $\Rightarrow$ $x=4$	check in the original تحقق في المعادلة الأصلية

Watch out انتبه

- $x=4$ may solve the new equation but not the original: always check.
- Multiply EVERY term, including the one on the right-hand side.

قد يحقق $x=4$ المعادلة الجديدة دون الأصلية: تحقق دائماً.

اضرب كل الحدود، ومنها الحد في الطرف الأيمن.

Now you try جرّب بنفسك

solutions at the back الحلول في آخر الكتيّب

1. Solve $\dfrac{x}{x-2} - 2 = \dfrac{2}{x-2}$.
2. Solve $\dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{1}{2}$.
3. Solve $\dfrac{x^2}{x-4} = \dfrac{16}{x-4} + 3$.

1 Solving rational equations

حلّ المعادلات النسبية

MAT.2.02.24.001

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 10 | MCQ · on the computer | اختيار من متعدد · على الحاسوب | Example 2 · p. 356

Example 2 Solve a Rational Equation with an Extraneous Solution

Solve $\frac{2m}{m-4} - \frac{m^2+7m+4}{3m^2-18m+24} = \frac{4m}{3m-6}$.

The LCD for the terms is $(m-4)(3m-6)$.

$$\frac{2m}{m-4} - \frac{m^2+7m+4}{3m^2-18m+24} = \frac{4m}{3m-6}$$

$$\frac{(m-4)(3m-6)(2m)}{(m-4)} - \frac{(m-4)(3m-6)(m^2+7m+4)}{3m^2-18m+24} = \frac{(m-4)(3m-6)(4m)}{3m-6}$$

$$\frac{1}{(m-4)(3m-6)(2m)} - \frac{1}{(m-4)(3m-6)(m^2+7m+4)} = \frac{1}{(m-4)(3m-6)(4m)}$$

$$(3m-6)(2m) - (m^2+7m+4) = (m-4)4m$$

$$6m^2 - 12m - m^2 - 7m - 4 = 4m - 16m$$

$$5m^2 - 19m - 4 = 4m - 16m$$

$$m^2 - 3m - 4 = 0$$

$$(m-4)(m+1) = 0$$

$$m-4 = 0 \text{ or } m+1 = 0$$

$$m = 4 \qquad m = -1$$

When solving a rational equation, any possible solution that results in a zero in the denominator must be excluded from the list of solutions.

Check each solution by substituting into the original equation.

Since $m = 4$ results in a zero in the denominator, it is extraneous. So, the solution is -1.

Check

Solve $\frac{3x}{x-4} - \frac{x^2-7x-4}{x^2-16} = \frac{5}{x+4}$.

$x =$?

2 Types of studies

أنواع الدراسات

Lesson 8-1 · Part 2 · pp. 372–376
MAT.4.02.09.001

A survey asks questions; an experiment applies a treatment to one group and compares; an observational study only observes.

المسح يطرح أسئلة، والتجربة تطبّق معالجة على مجموعة وتقارن، والدراسة الملاحظية تراقب فقط.

Steps الخطوات

1	**Are the people asked questions? → a survey**	يُسألون ← مسح
2	**Treatment on one group, then compared? → an experiment**	معالجة ومقارنة ← تجربة
3	**Only watched or measured? → an observational study**	مراقبة فقط ← دراسة ملاحظية
4	**Write the reason in one sentence**	اكتب السبب في جملة

Worked example مثال محلول

Researchers divide 60 volunteers into two groups: one takes a vitamin every day, the other none. The colds in each group are compared. Survey, experiment, or observational study?

the volunteers are divided into two groups — an experimental group and a control group / مجموعة تجريبية ومجموعة ضابطة

the researchers give one group a treatment (the vitamin) — the researchers apply a treat / لباحثون معالجة

the two groups are compared — the effects are compared / تُقارَن النتائج

an experiment — Learn: Types of Studies / Learn: أنواع الدراسات

Watch out انتبه

- An experiment needs a TREATMENT that the researcher applies. — التجربة تحتاج إلى معالجة يطبّقها الباحث.
- Comparing two groups is not enough to make it an experiment. — مقارنة مجموعتين لا تكفي لتكون تجربة.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. A school mails a questionnaire to 200 randomly selected students about their lunch. Survey, experiment, or observational study?
2. A scientist counts the birds visiting a park each morning for a month, without disturbing them. Survey, experiment, or observational study?
3. A teacher gives a new method to half of her classes and compares the results of the two halves. Survey, experiment, or observational study?

2 Types of studies

أنواع الدراسات

Lesson 8-1 · Part 2 · pp. 372–376
MAT.4.02.09.001

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 11 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Learn · Types of Studies · p. 372

Learn Types of Studies

In a **survey**, data are collected from responses given by members of a group regarding their characteristics, behaviors, or opinions.

In an **experiment**, the sample is divided into two randomly selected groups, the *experimental group* and the *control group*. The effect on the experimental group is then compared to the control group.

In an **observational study**, members of a sample are measured or observed without being affected by the study.

To determine when to use each type of study, think about how the data will be obtained and how the study will affect the participants. Each method of study relies on random selection of a sample to ensure that it accurately represents the population.

3 Theoretical and experimental probability
الاحتمال النظري والتجريبي

Lesson 8-2 · Part 1 · pp. 379–380
MAT.4.03.01.003

The theoretical probability is the number of favourable outcomes ÷ all the equally likely outcomes. The experimental probability is the number of times it happened ÷ the number of trials.

الاحتمال النظري = النتائج المرغوبة ÷ كل النتائج المتساوية الاحتمال. والاحتمال التجريبي = عدد مرات الحدوث ÷ عدد المحاولات.

Steps الخطوات

1	**Theory: 1 side out of all the sides of a fair die**	النظري: وجه واحد من كل أوجه النرد العادل
2	**Experiment: write count ÷ number of trials**	التجريبي: العدد ÷ عدد المحاولات
3	**Change each fraction to a percent**	حوّل كل كسر إلى نسبة مئوية
4	**Round only at the end (nearest tenth if asked)**	قرّب في النهاية فقط

Worked example مثال محلول

A student tossed a fair six-sided die 80 times and rolled a 6 twenty times. Find the theoretical and experimental probabilities of rolling a 6.

a fair die has 6 equally likely sides; one of them is a 6 — think: 1 favourable side out of 6 / فكّر: وجه مرغوب واحد من 6 أوجه

$$\text{theoretical} = \frac{1}{6} \approx 16.7\%$$ — theory = favourable ÷ all outcomes / النظري = المرغوبة ÷ كل النتائج

$$\text{experimental} = \frac{20}{80} = 25\%$$ — experiment = count ÷ trials / التجريبي = العدد ÷ المحاولات

Watch out انتبه

- The number of times it happened is a COUNT, not a probability: divide it by the trials.
- Theory and experiment are different numbers: do not swap them.

عدد مرات الحدوث عدد وليس احتمالًا: اقسمه على عدد المحاولات.

النظري والتجريبي عددان مختلفان: لا تبدّلهما.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. A student tossed a fair ten-sided die (numbered 1 to 10) 250 times and rolled a 7 thirty times. Find the theoretical and experimental probabilities of rolling a 7.
2. A student tossed a fair die 150 times and rolled a 2 sixteen times. Find the theoretical and experimental probabilities of rolling a 2. Round to the nearest tenth of a percent.
3. A student tossed a fair twelve-sided die 300 times and rolled a 12 twenty-one times. Find the theoretical and experimental probabilities of rolling a 12. Round to the nearest tenth of a percent.

3 Theoretical and experimental probability
الاحتمال النظري والتجريبي

Lesson 8-2 · Part 1 · pp. 379–380
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From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 12 | MCQ · on the computer | اختيار من متعدد · على الحاسوب | Example 1 · p. 379

Example 1 Find Probabilities

A student tossed a fair eight-sided die 200 times and recorded the results. Find the theoretical and experimental probabilities of rolling an 8.

Number on Die	Frequency
1	28
2	19
3	24
4	22
5	21
6	18
7	26
8	42

Theoretical Probability

The theoretical probability is what is expected to happen. Because the die is fair, each side has a $\frac{1}{8}$ chance of being the result. Thus, the theoretical probability is $\frac{1}{8}$, or 12.5%.

Experimental Probability

The experimental probability is based on the data collected from the experiment. Because the die was thrown 200 times and 42 of those throws landed on 8, the experimental probability is $\frac{42}{200}$, or 21%.

4 Describing a distribution: standard deviation
وصف التوزيع: الانحراف المعياري

Lesson 8-3 · pp. 388–391
MAT.4.02.04.010

The standard deviation shows how far the data are from the mean.

الانحراف المعياري يبيّن بُعد البيانات عن الوسط.

Steps الخطوات

1	**Step 1: find the mean μ**	الخطوة 1: أوجد الوسط μ
2	**Step 2: square each difference** $(\mu - x)^2$	الخطوة 2: ربّع كل فرق
3	**Step 3: add the squares · Step 4: divide by the number of values (the variance)**	الخطوة 3: اجمع · الخطوة 4: اقسم على عدد القيم (التباين)
4	**Step 5: take the square root (the standard deviation)**	الخطوة 5: خذ الجذر التربيعي (الانحراف المعياري)

Worked example مثال محلول

A teacher recorded the times (minutes) that 8 students took to finish a task: 2, 4, 4, 4, 5, 5, 7, 9. Find the standard deviation of the data.

$\mu = (2+4+4+4+5+5+7+9) \div 8 = 5$	the mean of the 8 values الوسط الحسابي للقيم الثماني
$9, 1, 1, 1, 0, 0, 4, 16$	the squares of (5 − each value) مربعات (5 − كل قيمة)
$9+1+1+1+0+0+4+16 = 32$	add the squares اجمع المربعات
$32 \div 8 = 4$ (the variance)	divide by the number of values: 8 اقسم على عدد القيم: 8
$\sqrt{4} = 2$ (the standard deviation)	take the square root خذ الجذر التربيعي

Watch out انتبه

- Divide by the number of values (n), not $n - 1$.
- The variance is not the answer: take the square root.

اقسم على عدد القيم (n) وليس $n - 1$.

التباين ليس الجواب: خذ الجذر التربيعي.

Now you try جرّب بنفسك solutions at the back الحلول في آخر الكتيّب

1. Five prices (dollars): 12, 15, 9, 20, 14. Find the standard deviation. Round to the nearest hundredth.
2. Books read each month: 3, 5, 8, 4, 6, 2, 7, 5. Find the standard deviation. Round to the nearest tenth.
3. Times of six runners (s): 10, 12, 14, 16, 18, 20. Find the standard deviation. Round to the nearest tenth.

4 Describing a distribution: standard deviation

وصف التوزيع: الانحراف المعياري

Lesson 8-3 · pp. 388–391

MAT.4.02.04.010

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 14 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Example 1 · p. 388 · Exercises 1, 2 · p. 391

Example 1 Find a Standard Deviation

TRACK **A coach recorded the times of each track member for a 400-meter race. Find and interpret the standard deviation of the data.**

400m Race Times (seconds)	
57.1	55.9
59.3	54.9
54.6	50.3
55.2	53.5

Step 1 Find the mean, μ.

$$\mu = \frac{57.1 + 59.3 + 54.6 + 55.2 + 55.9 + 54.9 + 50.3 + 53.5}{8}$$
$$= 55.1$$

The mean running time for the team is 55.1 seconds.

Step 2 Find the squares of the differences, $(\mu - x_n)^2$.

$(55.1 - 57.1)^2 = 4.00$ $(55.1 - 55.9)^2 = 0.64$

$(55.1 - 59.3)^2 = 17.64$ $(55.1 - 54.9)^2 = 0.04$

$(55.1 - 54.6)^2 = 0.25$ $(55.1 - 50.3)^2 = 23.04$

$(55.1 - 55.2)^2 = 0.01$ $(55.1 - 53.5)^2 = 2.56$

Step 3 Find the sum.

Find the sum of the values from **Step 2**.

$4.00 + 17.64 + 0.25 + 0.01 + 0.64 + 0.04 + 23.04 + 2.56$
$= 48.18$

Step 4 Divide by the number of values.

Divide the sum from **Step 3** by the number of running times.

$\frac{48.18}{8} = 6.0225$ This is the variance.

Step 5 Take the square root of the variance.

$\sqrt{6.0225} \approx 2.45$ This is the standard deviation.

The standard deviation is about 2.45. This is small compared with the run times, which means that the majority of the team members' times are close to the mean of 55.1 seconds, and almost all of the times will likely fall within 2 standard deviations of the mean, or between 50.2 and 60.0 seconds.

Check

How do you know that your solution is reasonable? (*Hint:* The majority of the data should be within one standard deviation of the mean, and almost all of the data should be within two.)

Over half of the times are within one standard deviation of the mean, which is the interval [52.65, 57.55] All of the data are within two standard deviations of the mean, or the interval [50.2, 60].

Example 1

1. BARBER A barber wants to purchase new professional shears from a Web site. The prices of all of the shears are shown in the table. Use the standard deviation formula to find and interpret the standard deviation of the data. Round your answers to the nearest cent.

Cost of Shears ($)			
50	165	55	79
84	68	38	42

2. READING Ms. Sanchez keeps track of the total number of books each student in the book club reads during the school year. Use the standard deviation formula to find and interpret the standard deviation of the data. Round your answers to the nearest tenth.

Books Read		
9	6	12
8	9	14
10	13	8

Solutions

written out step by step · الحلول

1 · Solving rational equations — حلّ المعادلات النسبية

1.1

Solve $\dfrac{x}{x-2} - 2 = \dfrac{2}{x-2}$.

$x - 2(x-2) = 2 \;\Rightarrow\; -x+4 = 2 \;\Rightarrow\; x = 2$ — multiply every term by $x-2$ / اضرب كل حد في $x-2$

check in the original / تحقق في المعادلة الأصلية

$x = 2$ makes a denominator zero; extraneous $\Rightarrow$ no solution

1.2

Solve $\dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{1}{2}$.

$2(x+3) + 2x = x(x+3) \;\Rightarrow\; 4x+6 = x^2+3x$ — LCD $2x(x+3)$ / المضاعف المشترك الأصغر $2x(x+3)$

$x^2 - x - 6 = 0 \;\Rightarrow\; (x-3)(x+2) = 0$ — factor / حلّل

$x = 3$ or $x = -2$ (neither makes a denominator zero) — check both / تحقق من الحلين

1.3

Solve $\dfrac{x^2}{x-4} = \dfrac{16}{x-4} + 3$.

$x^2 = 16 + 3(x-4) \;\Rightarrow\; x^2 - 3x - 4 = 0$ — multiply every term by $x-4$ / اضرب كل حد في $x-4$

$(x-4)(x+1) = 0 \;\Rightarrow\; x = 4$ or $x = -1$ — factor / حلّل

$x = 4$ is extraneous $\Rightarrow$ $x = -1$ — check in the original / تحقق في المعادلة الأصلية

2 · Types of studies — أنواع الدراسات

2.1

A school mails a questionnaire to 200 randomly selected students about their lunch. Survey, experiment, or observational study?

a questionnaire asks the students questions — the students answer questions / الطلاب يجيبون عن أسئلة

a survey — nothing is done to them / لا يُفعل بهم شيء

2.2

A scientist counts the birds visiting a park each morning for a month, without disturbing them. Survey, experiment, or observational study?

the birds are only watched and counted

no treatment, no questions
لا معالجة ولا أسئلة

an observational study

the sample is not affected
العينة لا تتأثر

2.3

A teacher gives a new method to half of her classes and compares the results of the two halves. Survey, experiment, or observational study?

a treatment is applied
تُطبَّق معالجة

the classes are divided into two groups; one group gets the new method (a treatment)

the results are compared: an experiment

two groups are compared
تُقارَن مجموعتان

3 · Theoretical and experimental probability — الاحتمال النظري والتجريبي

3.1

A student tossed a fair ten-sided die (numbered 1 to 10) 250 times and rolled a 7 thirty times. Find the theoretical and experimental probabilities of rolling a 7.

$\text{theoretical} = \frac{1}{10} = 10\%$

1 favourable side out of 10
وجه مرغوب واحد من 10 أوجه

$\text{experimental} = \frac{30}{250} = 12\%$

count ÷ trials
العدد ÷ المحاولات

3.2

A student tossed a fair die 150 times and rolled a 2 sixteen times. Find the theoretical and experimental probabilities of rolling a 2. Round to the nearest tenth of a percent.

$\text{theoretical} = \frac{1}{6} \approx 16.7\%$

1 favourable side out of 6
وجه مرغوب واحد من 6 أوجه

$\text{experimental} = \frac{16}{150} \approx 10.7\%$

count ÷ trials, rounded at the end
العدد ÷ المحاولات ثم التقريب في النهاية

3.3

A student tossed a fair twelve-sided die 300 times and rolled a 12 twenty-one times. Find the theoretical and experimental probabilities of rolling a 12. Round to the nearest tenth of a percent.

$\text{theoretical} = \frac{1}{12} \approx 8.3\%$

1 favourable side out of 12
وجه مرغوب واحد من 12 وجهًا

$\text{experimental} = \frac{21}{300} = 7\%$

count ÷ trials
العدد ÷ المحاولات

4.1

Five prices (dollars): 12, 15, 9, 20, 14. Find the standard deviation. Round to the nearest hundredth.

$\mu = \frac{12+15+9+20+14}{5} = 14$ — the mean / الوسط الحسابي

4, 1, 25, 36, 0 — square each difference / ربّع كل فرق

$4+1+25+36+0 = 66$ — add the squares / اجمع المربعات

$\frac{66}{5} = 13.2$ — divide by 5 values / اقسم على 5 قيم

$\sqrt{13.2} \approx 3.63$ — square root / الجذر التربيعي

4.2

Books read each month: 3, 5, 8, 4, 6, 2, 7, 5. Find the standard deviation. Round to the nearest tenth.

$\mu = \frac{3+5+8+4+6+2+7+5}{8} = 5$ — the mean / الوسط الحسابي

4, 0, 9, 1, 1, 9, 4, 0 — square each difference / ربّع كل فرق

$4+0+9+1+1+9+4+0 = 28$ — add the squares / اجمع المربعات

$\frac{28}{8} = 3.5$ — divide by 8 values / اقسم على 8 قيم

$\sqrt{3.5} \approx 1.9$ — square root / الجذر التربيعي

4.3

Times of six runners (s): 10, 12, 14, 16, 18, 20. Find the standard deviation. Round to the nearest tenth.

$\mu = \frac{10+12+14+16+18+20}{6} = 15$ — the mean / الوسط الحسابي

25, 9, 1, 1, 9, 25 — square each difference / ربّع كل فرق

$25+9+1+1+9+25 = 70$ — add the squares / اجمع المربعات

$\frac{70}{6} \approx 11.67$ — divide by 6 values / اقسم على 6 قيم

$\sqrt{11.67} \approx 3.4$ — square root / الجذر التربيعي